

Data Science

Chapter 1: Exploratory Data Analysis

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Principal Component
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What is Exploratory Data Analysis (EDA)?

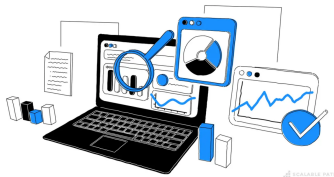
Before modeling, understand your data.

Definition

- Analyze datasets and summarize their main characteristics
- Use visualizations and descriptive statistics

Why is EDA important?

- Detect patterns and trends
- Identify outliers and missing values
- Check model assumptions
- Generate insights and hypotheses



Typical EDA techniques

- Data visualization (histograms, boxplots, scatter plots)
- Summary statistics (mean, median, variance)
- Correlation analysis

Key Message: *EDA transforms raw data into understanding and better decisions.*

Samples and Features in Tabular Data

ID	Age	Height (cm)	Math Score
1	15	168	85
2	16	172	78
3	15	165	92

- Each **row** represents one **observation** from the **sample**.
- Each **column** represents a **feature** (measured **variable**).
- The **population** is the complete group from which the sample was drawn (e.g., all students at Hochschule Kaiserslautern).

In machine learning, tabular data is often represented as a matrix:

$$X = \begin{bmatrix} x_{11} & \dots & x_{1d} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nd} \end{bmatrix} \in \mathbb{R}^{n \times d}$$

- n : number of **samples (observations)**
- d : number of **features (variables)**
- x_{ij} : value of feature j for sample i

Feature Types

- **Qualitative (categorical)**: Describe properties or categories, e.g.
 - Gender
 - Nationality
 - Legal form of a company
- **Quantitative**: Numerical values or measured quantities (including units), e.g.
 - Number of children
 - Annual income in EUR
 - Wind speed in m/s

Distinction:

- **Discrete**: Only specific, separated values are possible; obtained by **counting**, e.g.
 - Number of faces in a coin or die toss experiment
 - Number of people waiting in a checkout line
 - Number of 1 EUR coins in a wallet
- **Continuous**: Any real value within intervals is possible; obtained by **measurement**, e.g.
 - Body height in m
 - Deviation of a thread diameter from its target value in μm
 - Arrival time of a train at the station in hh:mm:ss

Note: Measurements inevitably involve rounding (measurement accuracy, resolution). However, the better the measuring instrument, the more precise the value (e.g., the number of meaningful decimal places). This applies to both analog and digital displays.

Measurement Levels

- **Nominal:** Only equality or difference can be identified; no ordering is possible; always qualitative, e.g.
 - Gender
 - Postal code
 - Nationality
- **Ordinal:** Categories are distinguishable and can be arranged in a natural or meaningful order, e.g.
 - School grade („very good“ to „insufficient“)
 - Intelligence quotient (IQ)
 - Ranking position in the Bundesliga
- **Metric (cardinal):** Expresses quantitative differences (not only order, but also distances); given as numbers/quantities (with units), e.g.
Distinction:
 - Temperature in K or $^{\circ}\text{C}$
 - Blood pressure in mmHG
 - Price in EUR
 - Running times (start/finish time) in h

Population

- The complete set of all individuals, objects, or observations of interest.
- It represents the entire group about which we want to draw conclusions.
- *Example:* All students at Hochschule Kaiserslautern, population of a country, . . .

Sample

- A subset of the population that is actually observed or measured.
- It is used to estimate or infer properties of the population.
- *Example:* 200 students randomly selected or considered in a specific course.

Example:

A car is selected as part of a drug-enforcement inspection sample. We speak of a **random sample** if, for example, every tenth car is checked and we assume that cars arrive in random order. A **systematic sample** would be checking all red vehicles. An **arbitrary sample** would occur if the officer selects vehicles without any criteria.

Observed feature values (Group a, Group b, Group a & b):

- Age **X** in years (quantitative, discrete, metric)
- Body height **Y** in cm (quantitative, continuous, metric)
- Vocational training **B** with values {yes, no} (qualitative, nominal)
- Gender **G** with values {m, w} (qualitative, nominal)

Merkmalsausprägung:	Alter [Jahre]	Min	19	Max	28
	Körperlänge [cm]	Min	155	Max	195
	Berufsausbildung [j/n]				
	Geschlecht [m/w]				

[illegible]

Urlisten, nach Zügen getrennt:

[illegible][illegible]

Frequency

Each feature value $\{x_1, \dots, x_i, \dots, x_k\}$ has the **absolute frequency**

$$n_i := \text{absFreq}(\mathbf{X} = x_i), \quad n_i > 0, \quad \sum_{i=1}^k n_i = n, \quad k \leq n$$

and the **relative frequency**

$$h_i := \text{relFreq}(\mathbf{X} = x_i) = \frac{n_i}{n}, \quad 0 < h_i \leq 1, \quad \sum_{i=1}^k h_i = 1, \quad k \leq n$$

Definition: Absolute and Relative Frequency

The tables

x_1	x_2	$\dots$	x_k
n_1	n_2	$\dots$	n_k

with $\sum_{i=1}^k n_i = n$

x_1	x_2	$\dots$	x_k
h_1	h_2	$\dots$	h_k

with $\sum_{i=1}^k h_i = 1$

are called the **absolute** and **relative frequency distributions**, respectively.

The frequency tables for the samples are:

[illegible]

For Group a & b combined, for example:

$$\mathbf{X} \in \{x_1 = 19, x_2 = 20, \dots, x_{10} = 28\}$$

$$\mathbf{Y} \in \{y_1 = 155, y_2 = 160, \dots, y_{27} = 195\}$$

$$\mathbf{B} \in \{b_1 = j, b_2 = n\}$$

$$\mathbf{G} \in \{g_1 = \text{m}, g_2 = \text{w}\}$$

$$\text{absFreq}(\mathbf{X} = x_2 = 20) = n_2 = 10$$

$$\text{relFreq}(\mathbf{Y} = y_4 = 163) = h_4 = \frac{3}{70}$$

$$\text{relFreq}(\mathbf{B} = b_1 = \text{j}) = h_1 = \frac{17}{70}$$

$$\text{absFreq}(\mathbf{G} = g_2 = \mathbf{w}) = n_2 = 23$$



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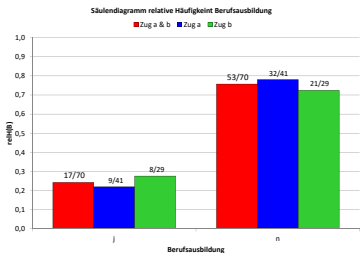
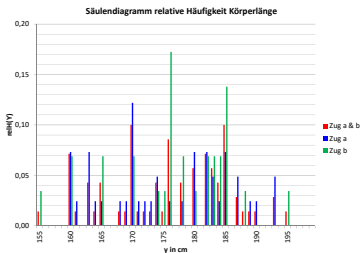
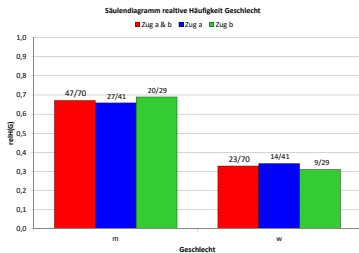
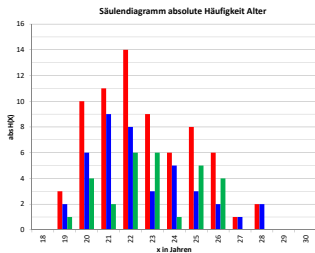
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Example: Sample from Statistics Course (WT 17/18)

Visualization of the features' frequencies as bar charts



Frequency & Distribution Function

For **metrically scaled feature values**, distances are defined, so that on a metric scale (typically on the horizontal axis / x -axis) the distances between feature values can be represented. On the vertical axis / y -axis, the corresponding frequencies can also be plotted metrically.

Definition: Frequency Function

The function

$$h(x) = \begin{cases} h_i & \text{if } x = x_i \\ 0 & \text{otherwise} \end{cases}$$

is called **frequency function** of the random variable $\mathbf{X}$.

Definition: Empirical Cumulative Distribution Function (ECDF)

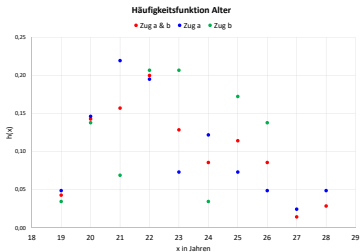
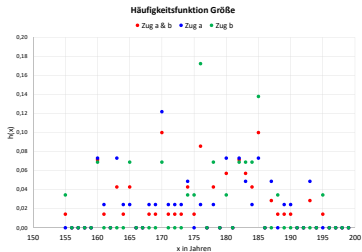
The function

$$H(x) = \sum_{x_i \leq x} h(x_i)$$

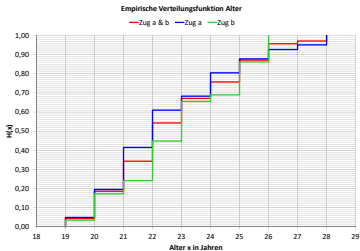
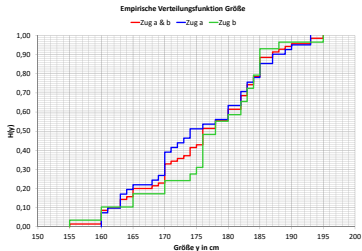
is called **empirical cumulative distribution function** of the random variable $\mathbf{X}$.

Example: Sample from Statistics Course (WT 17/18)

Frequency Functions



Empirical Cumulative Distribution Functions



Frequency & Distribution Function

The ECDF $H(x)$ has the following properties:

- 1 Everywhere at least right-continuous:

$$\lim_{\Delta x \rightarrow 0} H(x + \Delta x) = H(x), \quad \Delta x > 0$$

At the "jump points: Only right-continuous, i.e.

$$\lim_{\Delta x \rightarrow 0} H(x - \Delta x) \neq H(x), \quad \Delta x > 0$$

- 2 Monotonically increasing: $H(a) \leq H(b)$, if $a < b$, $a, b \in \mathbb{R}$

- 3 Limits:

$$\lim_{x \rightarrow -\infty} H(x) = 0, \quad \lim_{x \rightarrow +\infty} H(x) = 1$$

- 4 Difference: $H(b) - H(a) = \text{relFreq}(a < \mathbf{X} \leq b)$
- 5 Relationships between relative frequency and ECDF:

$$H(x) = \text{relFreq}(\mathbf{X} \leq x)$$

$$h(x) = H(x) - \lim_{\Delta x \rightarrow 0} H(x - \Delta x)$$

Earth Mover's Distance – Intuition

Intuitive idea:

- Two distributions = two piles of earth (or sand)
- Goal: reshape one pile into the other
- Minimize the required work

What defines work?

- How much mass is moved?
- How far is it moved?
- $\Rightarrow \text{Work} = \text{mass} \times \text{distance}$

Interpretation:

- Small EMD $\Rightarrow$ distributions are similar
- Large EMD $\Rightarrow$ distributions are very different

Also known as Wasserstein distance

Earth Mover's Distance – Formal Definition

Given:

- Two discrete distributions $P = (p_i)$ and $Q = (q_j)$
- Distance matrix $D = (d_{ij})$ with $d_{ij} = |p_i - q_j|$

Transport plan:

- γ_{ij} = mass moved from bin i to j
- Constraints:

$$\sum_j \gamma_{ij} = p_i$$

$$\sum_i \gamma_{ij} = q_j$$

$$\gamma_{ij} \geq 0$$

Definition:

$$EMD(P, Q) = \min_{\gamma} \sum_{i,j} \gamma_{ij} d_{ij}$$

Properties:

- Metric (under certain conditions)
- Sensitive to distribution structure
- Robust to small shifts

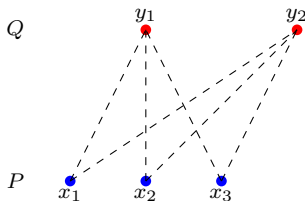
Earth Mover's Distance – Transport Plan Example

Distributions with 1D Histogram Bins

- $P = (0.3, 0.4, 0.3)$ with bins at positions $x = [1, 2, 3]$
- $Q = (0.5, 0.5)$ with bins at positions $y = [2, 4]$
- Distance: $d_{ij} = |x_i - y_j|$

Distance Matrix

$$D = \begin{pmatrix} 1 & 3 \\ 0 & 2 \\ 1 & 1 \end{pmatrix}$$



Each entry d_{ij} corresponds to the distance between x_i and y_j .

Earth Mover's Distance – Transport Plan Example

Optimal Transport Plan γ

$$\gamma = \begin{pmatrix} 0.3 & 0 \\ 0.2 & 0.2 \\ 0 & 0.3 \end{pmatrix}$$

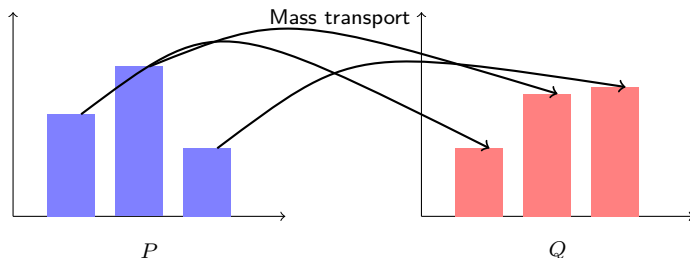
Interpretation:

- 0.3 from $x_1 \rightarrow y_1$
- 0.2 from $x_2 \rightarrow y_1$
- 0.2 from $x_2 \rightarrow y_2$
- 0.3 from $x_3 \rightarrow y_2$

Cost (EMD):

$$EMD = 0.3 \cdot 1 + 0.2 \cdot 0 + 0.2 \cdot 2 + 0.3 \cdot 1 = 1.0$$

Earth Mover's Distance – Visualization



Arrows indicate how probability mass is transported from P to Q .

Frequency Density & Histogram I

It is often useful to **condense** metric feature values into classes („classified data“), or the data may be presented in this form from the start.

Class Intervals

The interval of the real axis covered by the possible feature values is subdivided into m different sections (intervals) by appropriately chosen class boundaries

$$\xi_0 < \xi_1 < \xi_2 < \dots < \xi_m$$

with not necessarily equal class widths

$$\Delta_i = \xi_i - \xi_{i-1} > 0, \quad i = 1, \dots, m$$

Relative Frequency of Class Intervals

$$h_i := \text{relFreq}(\xi_{i-1} < \mathbf{X} \leq \xi_i), \quad i = 1, \dots, m$$

Relative Frequency Distribution of Class Intervals

- Assigned to class upper boundaries (common):

ξ_1	ξ_2	$\dots$	ξ_m
h_1	h_2	$\dots$	h_m

 with $\sum_{j=1}^m h_j = 1$

- Assigned to class midpoints (also possible):

$\frac{\xi_1 + \xi_0}{2}$	$\frac{\xi_2 + \xi_1}{2}$	$\dots$	$\frac{\xi_m + \xi_{m-1}}{2}$
h_1	h_2	$\dots$	h_m

 with $\sum_{j=1}^m h_j = 1$

- From the relative frequency distribution of the class intervals, one obtains the distribution function of the class intervals $H_C(x)$
- If the function values of $H_C(x)$ are connected at the jump points by straight lines (polygons), an approximating distribution function $\overline{H}(x)$ is obtained as a polygon.
- In addition to the polygon as an approximation, there are sometimes other, usually „smooth“ curves as „fits“, which can be calculated and displayed by the software used.

Frequency Density of Class Intervals

If $H_C(x)$ is the distribution function of a feature grouped into class intervals with class boundaries $\xi_0 < \xi_1 < \xi_2 < \dots < \xi_m$ and $\overline{H}(x)$ is the approximating distribution function represented by a polygon, then the quotient

$$\frac{H_C(\xi_i) - H_C(\xi_{i-1})}{\xi_i - \xi_{i-1}} = \frac{h_i}{\Delta_i}, \quad i = 1, \dots, m$$

is called the (average) **frequency density** of the i -th class interval. The first derivative

$$\overline{h}(x) = \frac{d\overline{H}(x)}{dx}$$

is called the **frequency density function** and its graph the **histogram**.

Remarks

- The histogram is a piecewise constant function (similar to a bar chart whose bars are adjacent to each other) that is defined for all values.
- In contrast to the graph of the frequency function, in a histogram it is not the height of the bar, but rather its area

$$\left(\frac{h_i}{\Delta_i}\right) \cdot \Delta_i$$

that indicates the relative frequency of the class interval!

- The total area of the histogram is always

$$\sum_{j=1}^m \left(\frac{h_j}{\Delta_j}\right) \cdot \Delta_j = 1$$

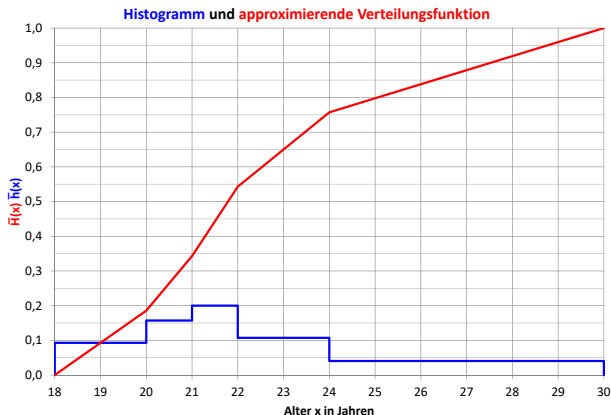
Example: Sample from Statistics Course (WT 17/18)

The feature value *age* can be condensed as follows, for example:

19	20	21	22	23	24	25	26	27	28
3	10	11	14	9	6	8	6	1	2



i	ξ_i	Δ_i	n_i	h_i	h_i/Δ_i	$H(\xi_i)$
0	18					0,000
1	20	2	13	13/70	39/420	0,186
2	21	1	11	11/70	66/420	0,343
3	22	1	14	14/70	84/420	0,543
4	24	2	15	15/70	45/420	0,757
5	30	6	17	17/70	17/420	1,000



Measures

A measure represents the most powerful form of condensing a statistical distribution into a single number.

Examples include:

- **Mean(s)**: Measure(s) for the „central location“ of the feature values („location measure“)
- **Dispersion measure(s)**: Measure(s) for the variation of feature values around the location measure
- **Quantile(s)**: A quantile is the feature value in a sorted dataset at which a specified percentage of all values is less than or equal to it

Arithmetic Mean

The arithmetic mean can be calculated for metrically scaled feature values.

Definition: Arithmetic Mean

The quantity

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j \quad \text{Raw data list}$$

$$= \frac{1}{n} \sum_{j=1}^k n_j \cdot x_j \quad \text{Abs. frequency}$$

$$= \sum_{j=1}^k h_j \cdot x_j \quad \text{Rel. frequency}$$

is called the **arithmetic mean** or **mean value** of a statistical distribution.

Remark

Strictly speaking, the overline $\bar{}$ is a kind of „operator notation“ in the sense that the result of averaging a quantity receives this overline. In this sense, $\overline{x^2}$ is the arithmetic mean of the squared feature values.

Example: Sample from Statistics Course (WT 17/18)

For Group a & b together it follows for the feature value *age* ($n = 70, k = 10$):

$$\begin{aligned}\bar{x}_{a \& b} &= \frac{1}{70} \sum_{j=1}^{70} x_j = \frac{1}{70} (22 + 21 + \dots + 22 + 23) = \frac{1.586}{70} \approx 22,657 \\ &= \frac{1}{70} \sum_{j=1}^{10} n_j \cdot x_j = \frac{1}{70} (3 \cdot 19 + 10 \cdot 20 + \dots + 2 \cdot 28) = \frac{1.586}{70} \approx 22,657 \\ &= \sum_{j=1}^{10} h_j \cdot x_j = \frac{3}{70} \cdot 19 + \frac{10}{70} \cdot 20 + \dots + \frac{2}{70} \cdot 28 = \frac{1.586}{70} \approx 22,657\end{aligned}$$

For both metric features *age* **X** and *body height* **Y** it follows:

Urliste gesamt (Zug a & b)			
Umfang	70 Elemente	Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste	22,657	Urliste	176,071
absH	22,657	absH	176,071
relH	22,657	relH	176,071

Urliste 1 (Zug a)			
Umfang	41 Elemente	Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste	22,488	Urliste	175,463
absH	22,488	absH	175,463
relH	22,488	relH	175,463

Urliste 2 (Zug b)			
Umfang	29 Elemente	Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste	22,897	Urliste	176,931
absH	22,897	absH	176,931
relH	22,897	relH	176,931

One can also compute the mean based on the means of the class intervals, e.g. ($k = 5$)

$$\bar{x} = \sum_{j=1}^5 h_j \cdot \bar{x}_j = \frac{13}{70} \cdot 19,769 + \dots + \frac{17}{70} \cdot 25,824 \approx 22,657$$

Note: Deviations may arise due to roundings.

i	ξ_i	Δ_i	n_i	h_i	$\bar{x}_i$
0	18				
1	20	2	13	13/70	19,769
2	21	1	11	11/70	21,000
3	22	1	14	14/70	22,000
4	24	2	15	15/70	23,400
5	30	6	17	17/70	25,824

Properties of the Arithmetic Mean

- ① Central property:

$$\sum_{j=1}^n (x_j - \bar{x}) = 0$$

- ② Shift by $a = \text{const.}$:

$$y_j = x_j + a \Rightarrow \bar{y} = \bar{x} + a$$

- ③ Scaling by $b = \text{const.}$:

$$z_j = b \cdot x_j \Rightarrow \bar{z} = b \cdot \bar{x}$$

- ④ Calculation from sub-series: The statistical series $\mathbf{X}$ is decomposed into m disjoint sub-series $\mathbf{X}_j$, $j = 1, \dots, m$, each with n_j elements and corresponding arithmetic means $\bar{x}_j$. Then we have

$$\bar{x} = \frac{1}{n} \sum_{j=1}^m n_j \cdot \bar{x}_j$$

e.g. $n = 70$, $m = 2$ (Group a & b), $n_1 = 41$, $n_2 = 29$:

$$\bar{x} = \frac{41}{70} \cdot 22,488 + \frac{29}{70} \cdot 22,897 \approx 22,656 \text{ (rounding inaccuracy)}$$

The median can be determined for metrically or ordinally scaled feature values.

Definition: Median

For a raw data list **sorted in ascending order**, the number x_{Med} with

$$\begin{aligned} x_{\text{Med}} &= x_{\frac{n+1}{2}} && n \text{ odd} \\ &= \frac{1}{2} (x_{\frac{n}{2}} + x_{\frac{n}{2}+1}) && n \text{ even and } \mathbf{X} \text{ metrically scaled} \\ x_{\frac{n}{2}} \leq x_{\text{Med}} \leq x_{\frac{n}{2}+1} &&& n \text{ even and } \mathbf{X} \text{ not metrically scaled} \end{aligned}$$

is called the **median** of the statistical series $\mathbf{X}$.

The median divides a raw data list sorted in ascending order in the middle, i.e. an equal number of observations lies to the left and right of the median, so it represents the “50% value”.

For the features *age* and *body height* (Group a & b, $n = 70$), the following holds (see next slide):

$$\begin{aligned} x_{\text{Med}} &= \frac{1}{2} (x_{\frac{70}{2}} + x_{\frac{70}{2}+1}) = \frac{1}{2} (x_{35} + x_{36}) = \frac{1}{2} (22 + 22) = 22 \\ y_{\text{Med}} &= \frac{1}{2} (y_{\frac{70}{2}} + y_{\frac{70}{2}+1}) = \frac{1}{2} (y_{35} + y_{36}) = \frac{1}{2} (176 + 176) = 176 \end{aligned}$$

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[illegible]

Definition: Variance and Standard Deviation

The mean squared deviation from the arithmetic mean $\bar{x}$

$$s_X^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x})^2$$

Raw data list

$$= \frac{1}{n} \sum_{j=1}^k n_j \cdot (x_j - \bar{x})^2$$

Absolute frequency

$$= \sum_{j=1}^k h_j \cdot (x_j - \bar{x})^2$$

Relative frequency

is called the **empirical variance**, or simply the **variance**, of the statistical series **X**. The positive root of s_X^2

$$s_X := +\sqrt{s_X^2}$$

is called the **empirical standard deviation**, or simply the **standard deviation**, of the statistical series **X**.

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Properties of Variance

- ① Non-negative:

$$s_X^2 \geq 0$$

- ② Shift by $a = \text{const.}$:

$$y_j = x_j + a \implies s_Y^2 = s_X^2$$

- ③ Scaling by $b = \text{const.}$:

$$z_j = b \cdot x_j \implies s_Z^2 = b^2 \cdot s_X^2 \text{ and } s_Z = |b| \cdot s_X$$

- ④ Simplified computation:

$$s_X^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x})^2 = \frac{1}{n} \left(\sum_{j=1}^n x_j^2 \right) - \bar{x}^2 = \overline{x^2} - \bar{x}^2$$

- ⑤ Computation from sub-series: The statistical series $\mathbf{X}$ is decomposed into m disjoint sub-series $\mathbf{X}_j$, $j = 1, \dots, m$, each with n_j elements and corresponding arithmetic means $\bar{x}_j$ and variances s_j^2 . Then

$$s_X^2 = \underbrace{\frac{1}{n} \sum_{j=1}^m n_j \cdot s_j^2}_{\text{Inner variance}} + \underbrace{\frac{1}{n} \sum_{j=1}^m n_j (\bar{x}_j - \bar{x})^2}_{\text{Outer variance}}$$

Example: Sample from Statistics Course (WT 17/18)

Variance calculations

Urliste gesamt (Zug a & b)			
Umfang 70 Elemente		Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste 22,657		Urliste 176,071	
absH 22,657		absH 176,071	
relH 22,657		relH 176,071	
Varianz aus		Varianz aus	
Urliste 4,940		Urliste 89,838	
absH 4,940		absH 89,838	
relH 4,940		relH 89,838	

Urliste 1 (Zug a)			
Umfang 41 Elemente		Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste 22,488		Urliste 175,463	
absH 22,488		absH 175,463	
relH 22,488		relH 175,463	
Varianz aus		Varianz aus	
Urliste 5,372		Urliste 94,005	
absH 5,372		absH 94,005	
relH 5,372		relH 94,005	

Urliste 2 (Zug b)			
Umfang 29 Elemente		Körperlänge	
Alter			
Mittelwert aus		Mittelwert aus	
Urliste 22,897		Urliste 176,931	
absH 22,897		absH 176,931	
relH 22,897		relH 176,931	
Varianz aus		Varianz aus	
Urliste 4,23		Urliste 82,685	
absH 4,23		absH 82,685	
relH 4,23		relH 82,685	

Combining outer and inner variance, $n = 70, m = 2$ (Group a & b),
 $n_1 = 41, n_2 = 29$, e.g. for *body height*:

$$\begin{aligned}
 s_Y^2 &= \underbrace{\frac{1}{70} (41 \cdot 94,005 + 29 \cdot 82,685)}_{\text{Inner variance, 89,315}} \\
 &+ \underbrace{\frac{1}{70} (41 \cdot (175,463 - 176,071)^2 + 29 \cdot (176,931 - 176,071)^2)}_{\text{Outer variance, 0,523}} \\
 &= 89,315 + 0,523 = 89,838
 \end{aligned}$$

Zusammenfassen Zug a & b					
Alter			Körperlänge		
Mittelwert		22,657	Mittelwert		176,071
Varianz innere		4,899	Varianz innere		89,315
äußere		0,041	äußere		0,523
Σ		4,940	Σ		89,838

Note on Variance: The Normalization Question

There is a **second definition of variance**, namely

$$s_X^2 = \frac{1}{n-1} \sum_{j=1}^n (x_j - \bar{x})^2$$

Raw data list

$$= \frac{1}{n-1} \sum_{j=1}^k n_j \cdot (x_j - \bar{x})^2$$

Absolute frequency

This definition normalizes by $n-1$ instead of n and provides, via s_X^2 , a „best possible“ estimate for the true, unknown variance of the unknown population from which the sample was taken.

For inferential statistics, this second definition is therefore the „better“ choice (unbiased point estimation).

However, both definitions can easily be converted into each other because they differ by the normalization ratio $\frac{n}{n-1}$.

Conclusion:

When reporting s_X^2 or s_X , it must be clear which normalization was used.

Further Dispersion Measures

There are additional dispersion measures that are not discussed further here:

- **Range:** Difference between the largest and smallest feature value:

$$\text{Range} := x_{\max} - x_{\min}$$

Applicable to metrically scaled feature values.

- **Mean absolute deviation:** Arithmetic mean of the absolute deviations of the feature values from the arithmetic mean $\bar{x}$

$$\text{MAD} := \frac{1}{n} \sum_{j=1}^n |x_j - \bar{x}|$$

Applicable to metrically scaled feature values.

- **Coefficient of variation:**

$$\text{VC}_X := \frac{s_X}{|\bar{x}|}, \quad \bar{x} \neq 0$$

Applicable to metrically scaled feature values.

Definition: Quantile

A number $x_{[q]}$ with $0 < q < 1$ is called the **q -quantile** if it partitions the statistical series $\mathbf{X}$ such that at least $100 \cdot q\%$ of its feature values are less than or equal to $x_{[q]}$, and at the same time at least $100 \cdot (1 - q)\%$ are greater than or equal to $x_{[q]}$, i.e.

$$\text{relFreq}(\mathbf{X} \leq x_{[q]}) \geq q \quad \text{and} \quad \text{relFreq}(\mathbf{X} \geq x_{[q]}) \geq 1 - q$$

- Typical, frequently used quantiles:

- Quartiles**

- Lower quartile: $Q_1 = x_{[0,25]}$
 - Middle quartile: $Q_2 = x_{[0,50]} = x_{\text{Med}}$ (median)
 - Upper quartile: $Q_3 = x_{[0,75]}$

From the quartiles, the **five-number summary** is obtained:

- Dividing the data set into the four sections

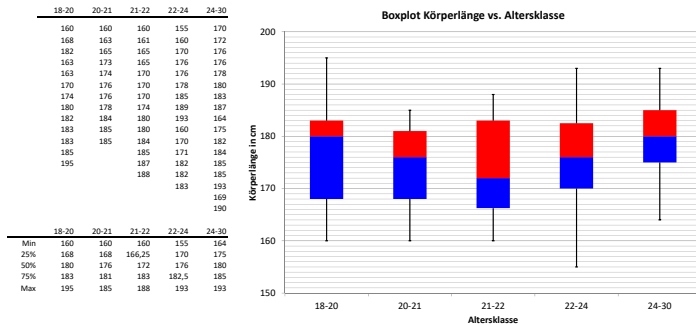
$$(x_{\min}, x_{[0,25]}, x_{\text{Med}}, x_{[0,75]}, x_{\max})$$

- Plotting them as a box plot (*Box-Plot*, *Box-Whisker-Plot*)

- Deciles:** $x_{[0,1]}, x_{[0,2]}, \dots, x_{[0,9]}$
 - Percentiles:** $x_{[0,01]}, x_{[0,02]}, \dots, x_{[0,99]}$

Example: Sample from Statistics Course (WT 17/18)

Box plots of *body height* vs. *age classes*



Note: The data are *not* sorted in ascending order.

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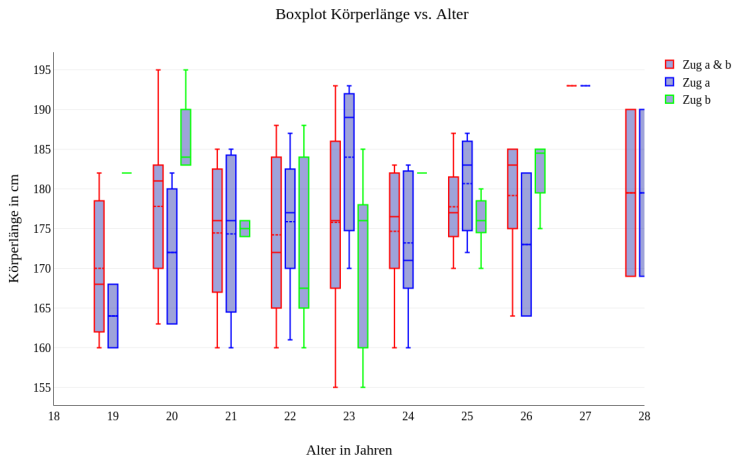
Abhängigkeitsmaße

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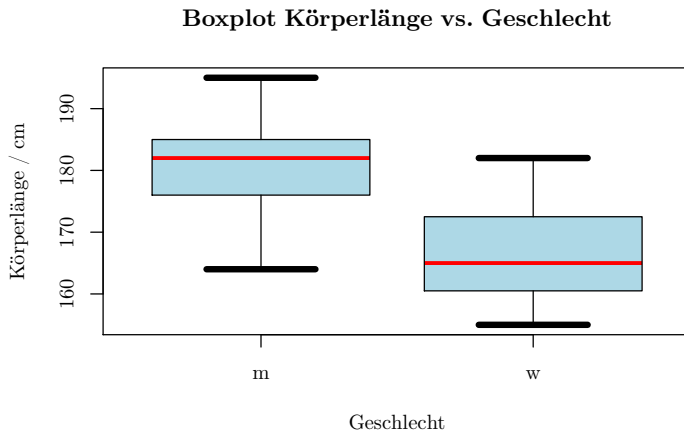
Example: Sample from Statistics Course (WT 17/18)

Box plots of *body height* vs. *age*, created with `http://plot.ly`



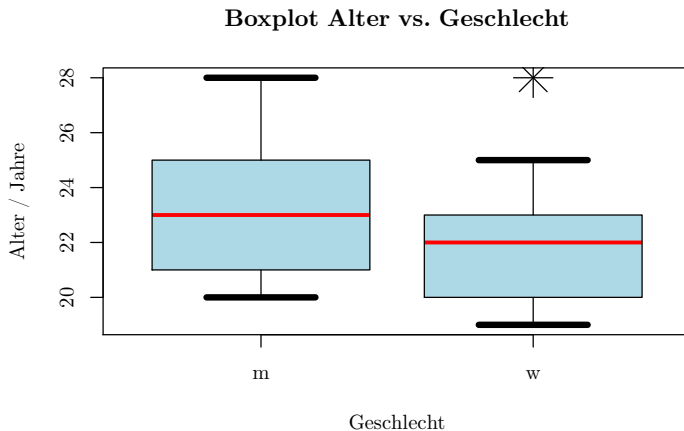
Example: Sample from Statistics Course (WT 17/18)

Box plots of *body height* vs. *gender*, created with R



Example: Sample from Statistics Course (WT 17/18)

Box plots of *age* vs. *gender*, created with R



Bisher: jedes Feature einzeln betrachtet ... Jetzt: Gemeinsame Eigenschaften unterschiedlicher Features ...

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Grundsätzliches

- Jede Zeile einer Datenmatrix ... Jede statistische Einheit ω_i , $i = 1, \dots, n$, einer Teil-/Grundgesamtheit Ω oder Stichprobe kann Träger mehrerer Merkmale $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_m$ sein, z.B.: Statistische Einheit oder Merkmalsträger: alle WI-Studierende, Merkmal $\mathbf{X}_1$: Körperlänge, Merkmal $\mathbf{X}_2$: Gewicht, Merkmal $\mathbf{X}_3$: Alter, ...
- Multivariate Statistik: Jede statistische Einheit ω_i , $i = 1, \dots, n$, einer Teil-/ Grundgesamtheit Ω oder Stichprobe kann Träger mehrerer Merkmale sein:

$$\mathbf{X}_1(\omega_i), \mathbf{X}_2(\omega_i), \dots, \mathbf{X}_m(\omega_i), \quad i = 1, 2, \dots, n$$

- Nachfolgend betrachtet: Zweidimensionale Statistik mit den zwei statistischen Variablen

$$\mathbf{X}(\omega_i), \mathbf{Y}(\omega_i), \quad i = 1, 2, \dots, n$$

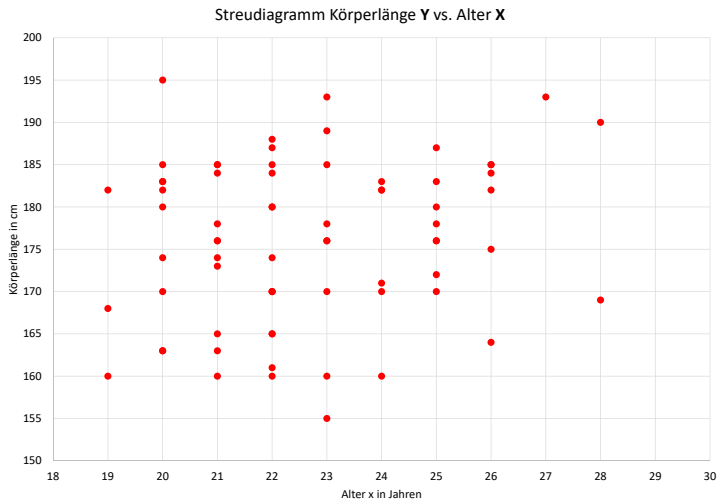
verkürzt $(\mathbf{X}, \mathbf{Y})$ geschrieben. Die beobachteten Merkmalsausprägungen sind $(\mathbf{X} = x_i, \mathbf{Y} = y_i)$, verkürzt (x_i, y_i) geschrieben.

Definition Streudiagramm

Im Streudiagramm werden die beobachteten Merkmalsausprägungen (x_i, y_i) als Punkte P_i interpretiert und in ein y - x -Koordinatensystem eingezeichnet.

Beispiel: Stichprobe in der Vorlesung Statistik im WS 1718

Für die Stichprobe ergibt sich z. B. (Zug a & b):



Kontingenz- oder Korrelationstabelle I

Definition Kontingenz- oder Korrelationstabelle

Die Kontingenz- oder Korrelationstabelle ist die Erweiterung eindimensionaler Häufigkeitstabellen auf zwei Dimensionen.

Absolute Häufigkeit, mit der die Wertekombination (x_i, y_j) entsprechend den gebildeten Größenklassen vorkommt:

$$\text{absFreq}(\mathbf{X} = x_i \cap \mathbf{Y} = y_j) = \text{absFreq}(x_i \cap y_j) = n_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, l$$

$\text{absFreq}(x_i \cap y_j)$	y_1	y_2	$\dots$	y_j	$\dots$	y_l	Zeilensummen
x_1	n_{11}	n_{12}	$\dots$	n_{1j}	$\dots$	n_{1l}	$n_{1\bullet}$
x_2	n_{21}	n_{22}	$\dots$	n_{2j}	$\dots$	n_{2l}	$n_{2\bullet}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_i	n_{i1}	n_{i2}	$\dots$	n_{ij}	$\dots$	n_{il}	$n_{i\bullet}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_k	n_{k1}	n_{k2}	$\dots$	n_{kj}	$\dots$	n_{kl}	$n_{k\bullet}$
Spaltensummen	$n_{\bullet 1}$	$n_{\bullet 2}$	$\dots$	$n_{\bullet j}$	$\dots$	$n_{\bullet l}$	n

Kontingenz- oder Korrelationstabelle II

- Absolute Häufigkeit, mit der x_i und y_j beobachtet wurden, aus Zeilen- bzw. Spaltensumme:

$$\text{absFreq}(\mathbf{X} = x_i \cap \mathbf{Y} \text{ beliebig}) = n_{i\bullet} = \sum_{j=1}^l n_{ij}, \quad i = 1, \dots, k$$

bzw.

$$\text{absFreq}(\mathbf{X} \text{ beliebig} \cap \mathbf{Y} = y_j) = n_{\bullet j} = \sum_{i=1}^k n_{ij}, \quad j = 1, \dots, l$$

- Gesamtanzahl an Beobachtungen:

$$n = \sum_{j=1}^l \sum_{i=1}^k n_{ij} = \sum_{i=1}^k n_{i\bullet} = \sum_{j=1}^l n_{\bullet j}$$

Kontingenz- oder Korrelationstabelle III

- Selbstverständlich kann die Kontingenztafel auch für die relativen Häufigkeiten erstellt werden, und dann gilt entsprechend:

$$\text{relFreq}(\mathbf{X} = x_i \cap \mathbf{Y} = y_j) = h_{ij} = \frac{n_{ij}}{n}, \quad i = 1, 2, \dots, k, \quad j = 1, \dots, l$$

$$\text{relFreq}(\mathbf{X} = x_i \cap \mathbf{Y} \text{ beliebig}) = h_{i\bullet} = \frac{n_{i\bullet}}{n} = \sum_{j=1}^l h_{ij}, \quad i = 1, \dots, k$$

$$\text{relFreq}(\mathbf{X} \text{ beliebig} \cap \mathbf{Y} = y_j) = h_{\bullet j} = \frac{n_{\bullet j}}{n} = \sum_{i=1}^k h_{ij}, \quad j = 1, \dots, l$$

$$1 = \sum_{j=1}^l \sum_{i=1}^k h_{ij} = \sum_{i=1}^k h_{i\bullet} = \sum_{j=1}^l h_{\bullet j}$$

Beispiel: Stichprobe in der Vorlesung Statistik im WS 1718

Für die Stichprobe ergibt sich z. B. (Zug a & b):

absH($x_i \cap y_j$)		Y =																											
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19		1					1										1											
	20				2					1				1				1	1	2		1						1	
	21		1		1		1						1	1		2	1				1	2							
	22		1	1			2			3				1				2			1	1	1	1					
	23	1	1							1						2	1								1				
	24		1							1	1								2	1			1				1		
	25										1		1								2	1							
	26					1										1	2	1	1		1		1						
	27															1				1								1	
	28								1																	1			

relH($x_i \cap y_j$)		Y =																											
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19		1/70					1/70											1/70										
	20				2/70					1/70				1/70				1/70	1/70	2/70		1/70						1/70	
	21		1/70		1/70		1/70						1/70	1/70		2/70	1/70				1/70	2/70							
	22		1/70	1/70			2/70			3/70				1/70				2/70				1/70	1/70	1/70					
	23	1/70	1/70							1/70						2/70	1/70					1/70	1/70	1/70		1/70			
	24		1/70							1/70	1/70								2/70	1/70		1/70	1/70			1/70			
	25									1/70		1/70									2/70	1/70							
	26					1/70									1/70					1/70		1/70	2/70						
	27																										1/70		
	28								1/70																	1/70			

Die $(k \times l)$ -Matrix ist eine (10×27) -Matrix, d. h. $i = 1, 2, \dots, 10$ und $j = 1, 2, \dots, 27$. Der Übersichtlichkeit halber sind keine Nullen eingetragen, z. B. ist der Eintrag $n_{53} = 0$ und daher leer.

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Definition Randverteilungen

Die eindimensionalen Verteilungen

$$\text{absFreq}(\mathbf{X} = x_i \cap \mathbf{Y} \text{ beliebig}) = n_{i\bullet} = \sum_{j=1}^l n_{ij}, \quad i = 1, 2, \dots, k$$

$$\text{absFreq}(\mathbf{X} \text{ beliebig} \cap \mathbf{Y} = y_j) = n_{\bullet j} = \sum_{i=1}^k n_{ij}, \quad j = 1, 2, \dots, l$$

$$\text{relFreq}(\mathbf{X} = x_i \cap \mathbf{Y} \text{ beliebig}) = h_{i\bullet} = \frac{n_{i\bullet}}{n} = \sum_{j=1}^l h_{ij}, \quad i = 1, 2, \dots, k$$

$$\text{relFreq}(\mathbf{X} \text{ beliebig} \cap \mathbf{Y} = y_j) = h_{\bullet j} = \frac{n_{\bullet j}}{n} = \sum_{i=1}^k h_{ij}, \quad j = 1, 2, \dots, l$$

heißen **Randverteilungen** der statistischen Variablen $\mathbf{X}$ bzw. $\mathbf{Y}$.

Anmerkung

Aus den Randverteilungen lassen sich natürlich arithmetisches Mittel $\bar{x}$ bzw. $\bar{y}$ und Varianz s_X^2 bzw. s_Y^2 errechnen.

Für die Stichprobe ergibt sich z. B. (Zug a & b):

absH($x_i \cap y_j$)		Y =																											
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19		1					1											1										3
	20				2					1				1				1	1	2		1						1	10
	21		1		1		1						1	1		2	1				1	2						11	
	22		1	1			2			3				1				2			1	1	1	1				14	
	23	1	1							1						2	1								1		1	9	
	24		1							1	1								2	1			1					8	
	25									1		1				2	1	1		1			1					8	
	26					1									1		1	1		1	2		1					6	
	27																									1	1	1	
28								1																	1	1	2	70	
		1	5	1	3	1	3	1	1	7	1	1	1	3	1	6	3	4	5	4	3	7	2	1	1	1	2	1	
		= n_j																											

relH($x_i \cap y_j$)		Y =																											
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19		1/70						1/70									1/70	1/70	2/70								1/70	3/70
	20				2/70					1/70					1/70			2/70	1/70	2/70		1/70	2/70					1/70	10/70
	21		1/70		1/70		1/70						1/70				2/70	1/70	1/70	2/70		1/70	2/70					1/70	11/70
	22		1/70	1/70			2/70			3/70				1/70			2/70				1/70	1/70	1/70	1/70				1/70	14/70
	23	1/70	1/70							1/70							2/70	1/70				1/70			1/70		1/70		9/70
	24		1/70							1/70	1/70						2/70	1/70	1/70		1/70					1/70		1/70	6/70
	25									1/70		1/70					2/70	1/70	1/70		1/70					1/70		1/70	8/70
	26					1/70							1/70		1/70		2/70	1/70	1/70		1/70	2/70					1/70	6/70	
	27																			1/70	1/70		1/70				1/70		1/70
	28								1/70																	1/70		1/70	2/70
		1/70	5/70	1/70	3/70	1/70	3/70	1/70	1/70	7/70	1/70	1/70	1/70	3/70	1/70	6/70	3/70	4/70	5/70	4/70	3/70	7/70	2/70	1/70	1/70	1/70	2/70	1/70	1,00
																						= h_1							

Die Randverteilungen sind in rot dargestellt.

Selbstverständlich sind diese identisch mit denen in Teil I, Kapitel 1 angegebenen Verteilungen für $\mathbf{X}$ und $\mathbf{Y}$.

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Bedingte Verteilungen

Interessiert man sich für die Verteilung eines Merkmals unter der **Bedingung**, dass das andere einen ganz bestimmten (festen) Wert hat, dann erhält man die sog. **bedingte Verteilung**, z. B. Verteilung der Körperlänge $\mathbf{Y}$ unter der Bedingung Alter $\mathbf{X} = 24$.

Schreibweise: Das Vorliegen der Bedingung wird durch einen senkrechten Strich, „|“, symbolisiert, z.B. $\mathbf{Y}|\mathbf{X} = 24$

Definition der bedingten Verteilung

Die $j = 1, 2, \dots, l$ eindimensionalen Verteilungen

$$\text{relFreq}(\mathbf{X} = x_i | \mathbf{Y} = y_j) = h_{i|y_j} = \frac{h_{ij}}{h_{\bullet j}} = \frac{n_{ij}}{n_{\bullet j}}, \quad i = 1, \dots, k$$

und die $i = 1, 2, \dots, k$ eindimensionalen Verteilungen

$$\text{relFreq}(\mathbf{Y} = y_j | \mathbf{X} = x_i) = h_{j|x_i} = \frac{h_{ij}}{h_{i\bullet}} = \frac{n_{ij}}{n_{i\bullet}}, \quad j = 1, \dots, l$$

heißen **bedingte Verteilungen**.

Motivation & Introduction

Für die Stichprobe ergibt sich z. B. (Zug a & b):

relH($y_j \mid x_i$)		Y =																												
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195		
X =	19		1/3					1/3										1/3	1/3	2/10	1/10							1/10	3/3	Σ
	20			2/10					1/10				1/10				1/10	1/10	2/10	1/10									10/10	
	21		1/11		1/11		1/11					1/11	1/11			2/11	1/11		1/11	2/11	1/11								11/11	
	22		1/14	1/14			2/14			3/14			1/14					2/14			1/14	1/14	1/14	1/14					14/14	
	23	1/9	1/9							1/9						2/9	1/9					1/9			1/9	1/9			9/9	
	24		1/6							1/6	1/6								2/6	1/6					1/9				6/6	
	25									1/8		1/8					2/8	1/8	1/8	1/8			1/8						8/8	
	26					1/6								1/6					1/6		1/6	2/6							6/6	
	27																									1/1			1/1	
28								1/2																	1/2			2/2		

relH(x_i, y_j)		Y =																											
		155	160	161	163	164	165	168	169	170	171	172	173	174	175	176	178	180	182	183	184	185	187	188	189	190	193	195	
X =	19		1/5					1/1											1/4										
	20				1/2					1/5				1/3				1/3	1/4	2/3		1/5						1/1	
	21		1/5		1/2		1/2						1/1				1/3	1/3		1/3	1/5	1/5							
	22		1/5	1/1			1/2			2/5				1/3				2/3			1/3	1/5	1/2	1/1					
	23	1/1	1/5							1/5							1/3	1/3				1/5	1/2	1/1			1/2		
	24		1/5							1/5	1/1									2/4	1/3		1/5			1/1			
	25									1/5		1/1					1/3	1/3		1/3	1/3		1/2						
	26				1/1											1/1		1/3	1/3	1/3	1/4	1/3	1/5						
	27																										1/2		
	28							1/1																		1/1	1/1	1/2	1/1
		1/1	5/5	1/1	2/2	1/1	2/2	1/1	1/1	5/5	1/1	1/1	1/1	3/3	1/1	3/3	3/3	3/3	4/4	3/3	3/3	5/5	2/2	1/1	1/1	1/1	2/2	1/1	
		Σ																											

Es gibt

- $k = 10$ bedingte Verteilungen $\text{relFreq}(\mathbf{Y} = y_j | \mathbf{X} = x_i)$ (oben)
- $l = 27$ bedingte Verteilungen $\text{relFreq}(\mathbf{X} = x_i | \mathbf{Y} = y_j)$ (unten)

Auffällig ist, dass die Wahl einer Bedingung für eine Variable anscheinend die Möglichkeiten der anderen Variablen einschränken kann, z. B. ist durch die Bedingung $\mathbf{Y} = 155$ die Variable $\mathbf{X}$ nicht nur eingeschränkt, sondern sogar eindeutig auf $\mathbf{X} = 23$ festgelegt.

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Definition statistischer Unabhängigkeit

Ist die gemeinsame Verteilung h_{ij} gleich dem Produkt der Randverteilungen, d. h. gilt

$$h_{ij} = h_{i\bullet} \cdot h_{\bullet j}, \quad i = 1, \dots, k, \quad j = 1, \dots, l$$

so heißen **X** und **Y** **statistisch unabhängig**.

Anmerkungen:

- 1 Bei statistischer Unabhängigkeit ist jeder Eintrag der Kontingenztabelle h_{ij} gleich dem Produkt aus Zeilensumme $h_{i\bullet}$ und Spaltensumme $h_{\bullet j}$:

$$h_{i|y_j} = \frac{h_{ij}}{h_{\bullet j}} = \frac{h_{i\bullet} \cdot h_{\bullet j}}{h_{\bullet j}} = h_{i\bullet}, \quad i = 1, 2, \dots, k$$

$$h_{j|x_i} = \frac{h_{ij}}{h_{i\bullet}} = \frac{h_{i\bullet} \cdot h_{\bullet j}}{h_{i\bullet}} = h_{\bullet j}, \quad j = 1, 2, \dots, l$$

- 2 Anschaulich heißt statistische Unabhängigkeit, dass die Ausprägung des Merkmals **X** keinen Rückschluss auf diejenige des Merkmals **Y** (und umgekehrt) erlaubt
- 3 Sind **X** und **Y** statistisch abhängig, dann sagt dies nichts über eine physikalische Abhängigkeit (Ursache–Wirkung) aus!

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Definition Kovarianz

Die Größe

$$\begin{aligned}c_{XY} &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) \cdot (y_i - \bar{y}) \\&= \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^l n_{ij} \cdot (x_i - \bar{x}) \cdot (y_j - \bar{y}) \\&= \sum_{i=1}^k \sum_{j=1}^l h_{ij} \cdot (x_i - \bar{x}) \cdot (y_j - \bar{y}) \\&= \bar{x} \cdot \bar{y} - \bar{x} \cdot \bar{y}\end{aligned}$$

heißt **empirische Kovarianz** oder kurz **Kovarianz** zwischen den statistischen Variablen **X** und **Y**.

Anmerkung:

Die Kovarianz c_{XY} wird null, sofern die statistischen Variablen **X** und **Y** unabhängig sind. Die Kovarianz c_{XY} sagt demnach etwas über die statistische Gemeinsamkeit zwischen **X** und **Y** aus! **Abbildung aus Schira**

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Step 1: Tabular Data

Height	Weight	Age
170	65	15
180	75	16
175	70	15

Step 2: Matrix Representation

$$X = \begin{bmatrix} 170 & 65 & 15 \\ 180 & 75 & 16 \\ 175 & 70 & 15 \end{bmatrix} \in \mathbb{R}^{3 \times 3}$$

Step 3: Center the Data

Compute the mean of each feature: $\mu = [175 \quad 70 \quad 15.33] \in \mathbb{R}^{1 \times 3}$

Center the data by subtracting the mean:

$$X_c = X - \begin{bmatrix} \mu \\ \mu \\ \mu \end{bmatrix} = \begin{bmatrix} -5 & -5 & -0.33 \\ 5 & 5 & 0.67 \\ 0 & 0 & -0.33 \end{bmatrix}$$

$\implies$ Centering ensures that each feature has mean 0

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Step 4: Covariance Matrix

Covariance matrix of centered data:

$$\Sigma = \frac{1}{n} X_c^T X_c \approx \begin{bmatrix} 16.67 & 16.67 & 0.67 \\ 16.67 & 16.67 & 0.67 \\ 0.67 & 0.67 & 0.33 \end{bmatrix}$$

- Variance of Height = 16.67
- Variance of Weight = 16.67
- Variance of Age = 0.33
- Covariance Height-Weight = 16.67 (strong correlation)
- Age contributes very little

Summary:

$$s_{\text{Height}}^2 = 16.67, s_{\text{Weight}}^2 = 16.67, s_{\text{Age}}^2 = 0.33$$

$$c_{\text{Height,Weight}} = 16.67, c_{\text{Height,Age}} = 0.67, c_{\text{Weight,Age}} = 0.67$$

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Step 5: Eigenvalues and Principal Components

Solve the eigenvalue problem $\Sigma v_i = \lambda_i v_i$:

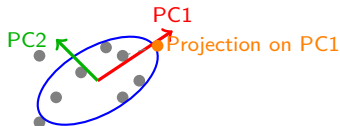
- $\lambda_1 \approx 33 \rightarrow \text{PC1}$ (largest variance)
- $\lambda_2 \approx 0.33 \rightarrow \text{PC2}$
- $\lambda_3 \approx 0 \rightarrow \text{PC3}$ (almost no variance)
- PC1 vector $\approx [0.71, 0.71, \varepsilon]$, ε small

Interpretation

Height and Weight dominate PC1 $\rightarrow$ almost all variance captured

Age has very little contribution $\rightarrow$ nearly PC3

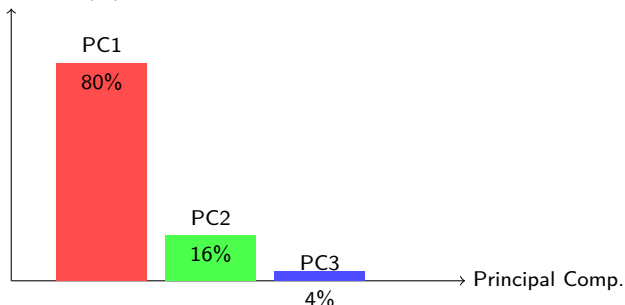
Step 6: Project Data onto Principal Components



- PC1: direction of **maximum variance** (red)
- PC2: orthogonal to PC1 (green)
- Projecting data onto PCs reduces dimensionality while retaining most variance

Explained Variance of Principal Components

Explained Variance (%)



- PC1, captures largest variation
- PC2, some variation, contributes moderately
- PC3, contributes very little, negligible
- Choosing first 1-2 PCs effectively reduces dimensionality
- This shows why PCA can **reduce dimensions without losing much information**

Definition Korrelationskoeffizient

Der Quotient

$$r_{XY} = \frac{c_{XY}}{s_X \cdot s_Y}, \quad s_X > 0, s_Y > 0$$

heißt **empirischer Korrelationskoeffizient** zwischen den statistischen Variablen **X** und **Y**.

Anmerkungen:

- ① Die Größe r_{XY} hat das gleiche Vorzeichen wie c_{XY} , so dass gilt

$$-1 \leq r_{XY} \leq +1$$

mit Interpretation

$$-1 \leq r_{XY} < 0 \quad \Leftrightarrow \quad \text{negative lineare Korrelation}$$

$$r_{XY} = 0 \quad \Leftrightarrow \quad \text{keine lineare Korrelation}$$

$$0 < r_{XY} \leq +1 \quad \Leftrightarrow \quad \text{positive lineare Korrelation}$$

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Metrische Variablen: Korrelationskoeffizient II

- 2 Maßstabsänderungen (genauer: lineare Transformationen) beeinflussen r_{XY} nicht
- 3 Vertauschen von $\mathbf{X}$ und $\mathbf{Y}$ verändert r_{XY} nicht, d. h. es gilt

$$r_{XY} = r_{YX}$$

- 4 r_{XY} gibt lediglich eine *statistische Tendenz* an. Eine *kausale Beziehung* (im Sinne von Ursache–Wirkung) zwischen den Merkmalen $\mathbf{X}$ und $\mathbf{Y}$ muss es deshalb nicht geben, sie könnte auch rein zufällig sein.
Beispiel:

Land	Anzahl Störche je 100 km ²	Anzahl Geburten je 1000 Einwohner
	X	Y
Deutschland	3,5	0,8
Frankreich	3,9	0,9
USA	4,2	1,1
Brasilien	7,1	1,4
Argentinien	6,9	1,3
Rußland	4,5	1,2
Ägypten	12,8	1,8
Namibia	10,5	1,5
China	13,2	1,6
Indonesien	10,2	1,4
Indien	12,5	1,9

$r_{XY} = 0,92$

- 5 r_{XY} kann nur für metrisch skalierte Merkmale $\mathbf{X}$ und $\mathbf{Y}$ errechnet werden.
- 6 r_{XY} wird oft als **Korrelationskoeffizient nach Breavais-Pearson** bezeichnet.

Beispiel: Stichprobe in der Vorlesung Statistik im WS 1718

Für die Stichprobe ergibt sich z. B. (Zug a & b) folgende Arbeitstabelle $x_i \cdot y_j \cdot h_{ij}$ zur Berechnung von $\overline{x \cdot y}$:

$x_i \cdot y_j \cdot h_{ij}$		$y =$													
		155	160	161	163	164	165	168	169	170	171	172	173	174	175
$X =$	19		3040/70					3192/70		3400/70				3480/70	
	20				6520/70									3654/70	
	21		3360/70		3423/70		3465/70						3633/70		
	22		3520/70	3542/70			7260/70			11220/70				3828/70	
	23	3565/70	3680/70							3910/70					
	24		3840/70							4080/70	4104/70				
	25									4250/70		4300/70			
	26					4264/70									4550/70
	27														
	28								4732/70						

$x_i \cdot y_j \cdot h_{ij}$		$y =$													
		176	178	180	182	183	184	185	187	188	189	190	193	195	
$X =$	19				3458/70										
	20			3600/70	3640/70	7320/70		3700/70						3900/70	
	21	7392/70	3738/70				3864/70	7770/70							
	22			7920/70			4048/70	4070/70	4114/70	4136/70					
	23	8096/70	4094/70				4255/70				4347/70		4439/70		
	24				8736/70	4392/70									
	25	8800/70	4450/70	4500/70		4575/70			4675/70						
	26				4732/70		4784/70	9620/70							
	27														
	28											5320/70	5211/70		

Alle Einträge summiert ergibt $\overline{x \cdot y} = \frac{279.508}{70}$. Mit $\bar{x} = \frac{1.586}{70}$ und $\bar{y} = \frac{12.325}{70}$ folgt $c_{XY} = \overline{x \cdot y} - \bar{x} \cdot \bar{y} = \frac{18.110}{4.900}$. Mit $\sigma_X^2 = \frac{24.204}{4.900}$ und $\sigma_Y^2 = \frac{440.205}{4.900}$ ist der Korrelationskoeffizient

$$r_{XY} = \frac{c_{XY}}{s_X \cdot s_Y} = \frac{18.110/4.900}{\sqrt{\frac{24.204}{4.900}} \sqrt{\frac{440.205}{4.900}}} = \frac{18.110}{\sqrt{24.204 \cdot 440.205}} \approx 0,175$$

was eine schwache positive Korrelation (einen schwachen positiven linearen Zusammenhang) bedeutet.

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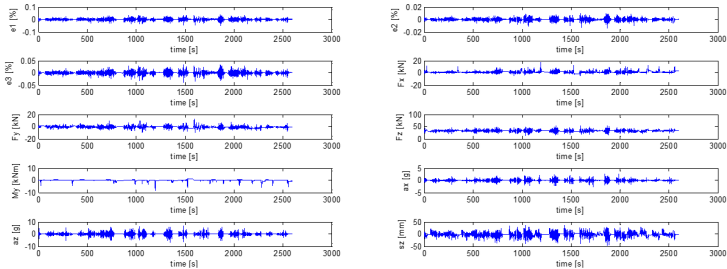
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Example: Load Data Analysis

Data example: Measurements in the area of the front wheel on a truck during a city trip

3 strains, 3 wheel forces, drive torque, 2 accelerations, spring displacement

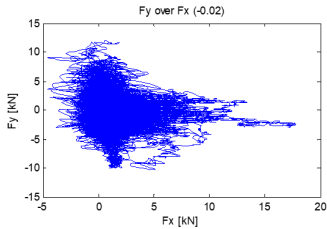


Open

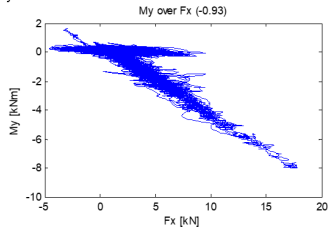
Example: Load Data Analysis

2D phase plots – a simple display option for visual evaluation

- **Fy over Fx** (very low negative correlation)
Lateral and longitudinal forces occur largely independently of each other.



- **My via Fx** (very high negative correlation)
The driving or braking torque leads to longitudinal forces. The longitudinal forces at **My = 0** are generated by unevenness.



Open

Example: Load Data Analysis

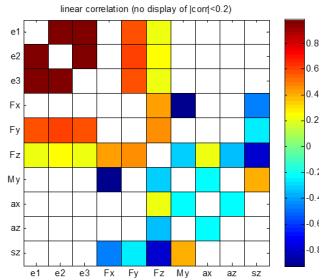
Linear (empirical) correlation coefficient – simple scalar variable for quantitative statements

Correlation between the variables $x_j(t)$, $x_k(t)$:

$$r_{j,k} = \frac{\sum_i (x_j(t_i) - \bar{x}_j) \cdot (x_k(t_i) - \bar{x}_k)}{\sqrt{\sum_i (x_j(t_i) - \bar{x}_j)^2 \cdot \sum_i (x_k(t_i) - \bar{x}_k)^2}}, \bar{x}_j = \frac{1}{N} \cdot \sum_i x_j(t_i)$$

Application to the sample data:

- Strains very highly correlated with each other
- Strains with Fy highly correlated with Fz lowly correlated
- Fz and sz strongly negatively correlated
- Fz slightly correlated with Fx and Fy
- My strongly negatively / positively correlated with Fx / sz



Open